

The Power of Vaccination: From Smallpox Eradication to COVID-19 in Record Time

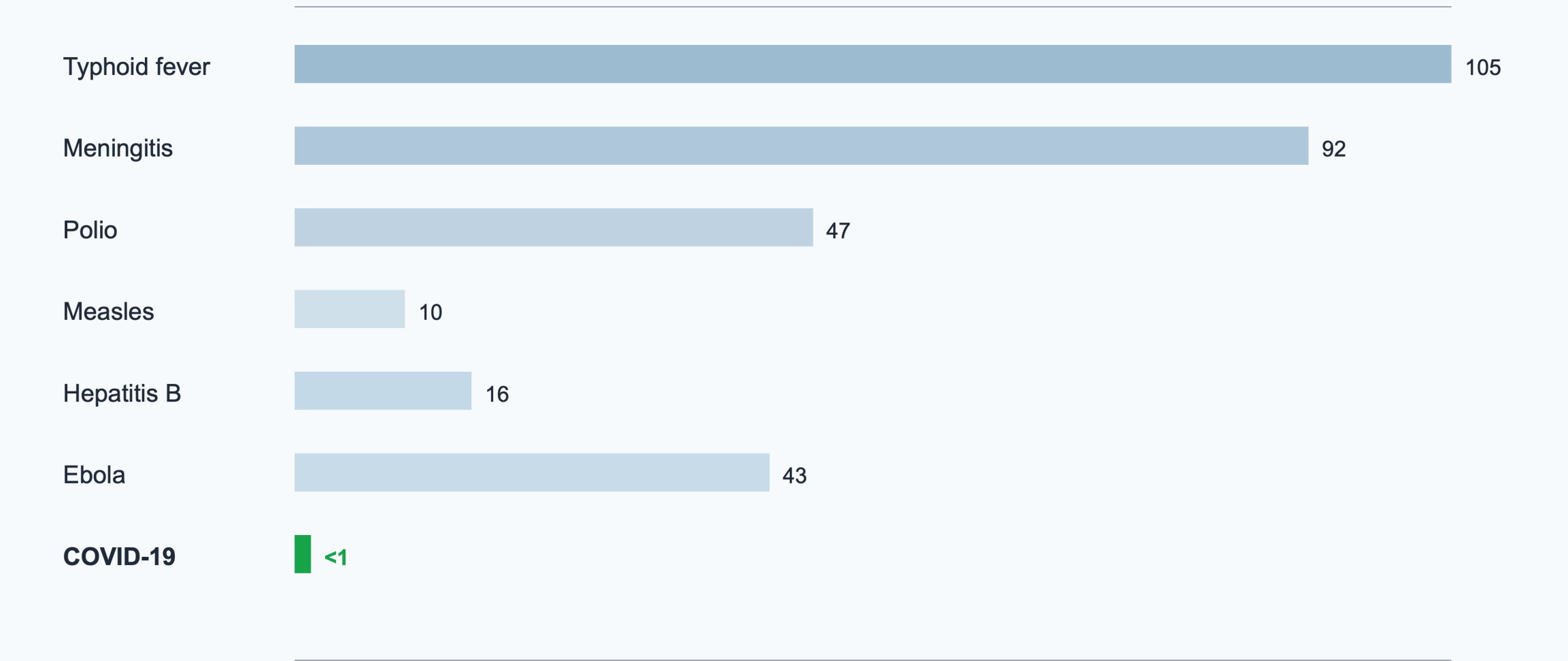
A data-driven overview of vaccination impact from 1796–2020

Audience: Public health professionals and policy analysts

Goal: Quantify progress, explain mechanisms, and identify coverage gaps

From 100 Years to Under 1 Year: Vaccine Development Is Accelerati

Gap between pathogen identification and vaccine licensure (years)



Insight: Development timelines collapsed from 105 years (typhoid) to under 1 year (COVID-19).

Source: Our World in Data table in source brief

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in t

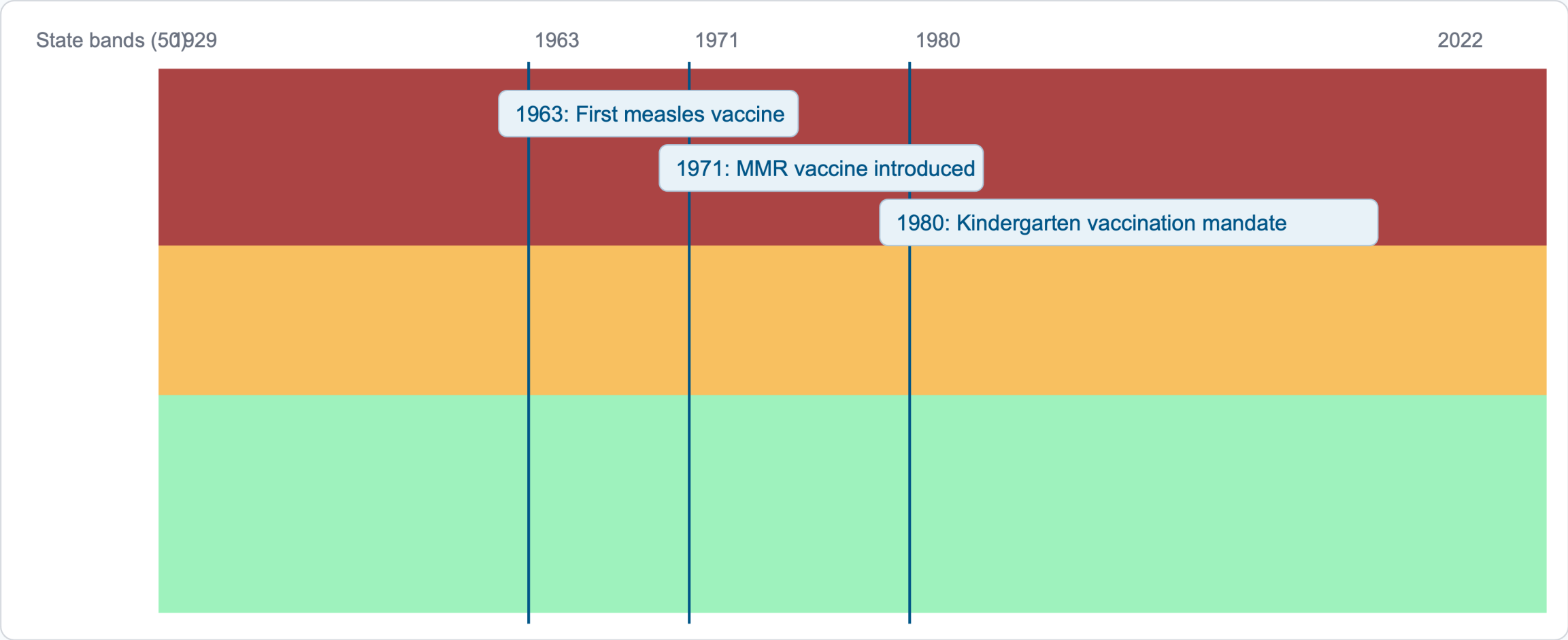
Disease	Pre-case	Post-case	Case reduction	Pre-death	Death reduction
Diphtheria	158	0	100%	13.7	100%
Measles	3,044	0.2	99.99%	2.5	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Green = 100% elimination; amber = remaining burden requiring policy focus.

Source: Roush & Murphy (2007) values as provided in source brief

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Heatmap-style representation of measles incidence trends, 1929–2022 (cases per 100,000)



Before 1963: high incidence (100–1,000+). After 1980: near zero across states, with sporadic outbreaks.

Source: Measles heatmap description and milestones from source brief

Smallpox: The Only Human Disease Eradicated Through Vaccination

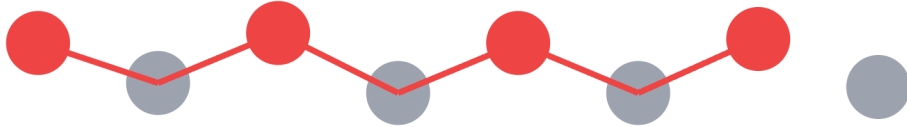


Why eradication was possible

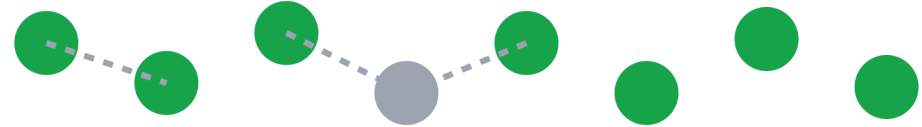
- Virus infected only humans (no animal reservoir)
- Symptoms were clinically distinguishable
- Vaccine was cheap and effective for mass campaigns

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Low coverage: transmission chains persist



High coverage: spread is interrupted



Grey nodes represent people who cannot be vaccinated (newborns, immunocompromised patients, chemotherapy patients).

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Incomplete disease reduction (US)

Pneumococcal: 40.5% case reduction, 31.3% death reduction



Hepatitis A: 89% case reduction



Acute hepatitis B: 83.9% case reduction



Barriers to close

- Uneven global access
- Supply shortages
- Lower trust and uptake

Policy implication:

Prioritize delivery systems, risk communication, and coverage-focused programs.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job

- 1) Vaccination achieved 100% elimination of cases and deaths for diphtheria, smallpox, and acute polio in the US.
- 2) Development timelines collapsed from 105 years (typhoid) to under 1 year (COVID-19).
- 3) State-level measles burden fell from 100–1,000+ to near zero after the 1980 mandate.
- 4) Herd immunity protects those who cannot be vaccinated.
- 5) Coverage and equity gaps remain for pneumococcal disease and hepatitis A/B.

Call to Action for Global Health Systems

Expand access, strengthen supply resilience, and rebuild trust to close the remaining vaccination gap.